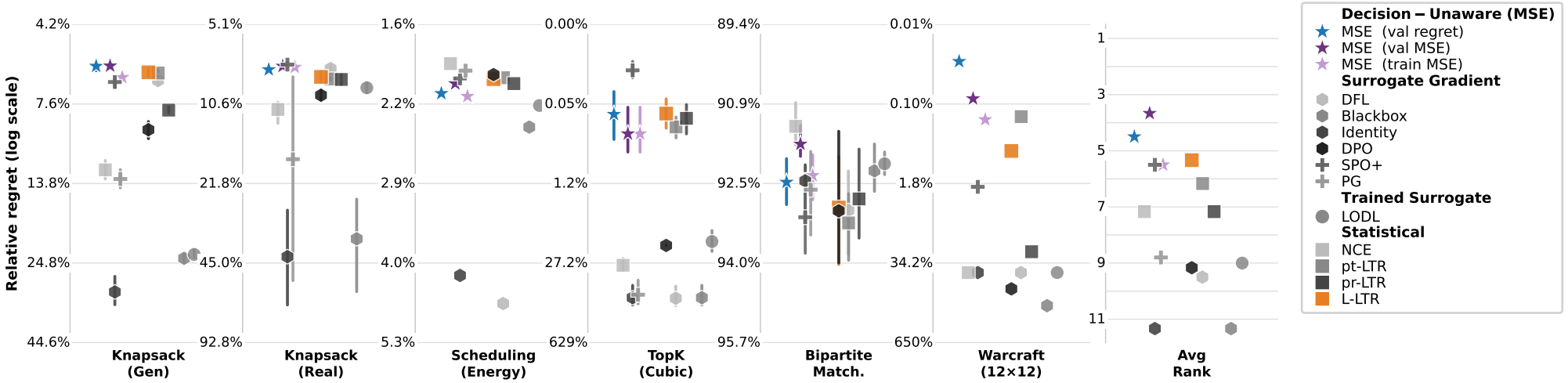
Benchmark re-run — 10-seed mean ± 95% bootstrap CI — Relative regret % (best ↑)
Relative regret (log scale)
4.2%
7.6%
13.8%
24.8%
44.6%
5.1%
10.6%
21.8%
45.0%
92.8%
1.6%
2.2%
2.9%
4.0%
5.3%
0.00%
0.05%
1.2%
27.2%
629%
89.4%
90.9%
92.5%
94.0%
95.7%
0.01%
0.10%
1.8%
34.2%
650%
1
3
5
7
9
11
Knapsack (Gen)
Knapsack (Real)
Scheduling (Energy)
TopK (Cubic)
Bipartite Match.
Warcraft (12×12)
Avg Rank
Decision – Unaware (MSE)
MSE (val regret)
MSE (val MSE)
MSE (train MSE)
Surrogate Gradient
DFL
Blackbox
Identity
DPO
SPO+
PG
Trained Surrogate
LODL
Statistical
NCE
pt-LTR
pr-LTR
L-LTR